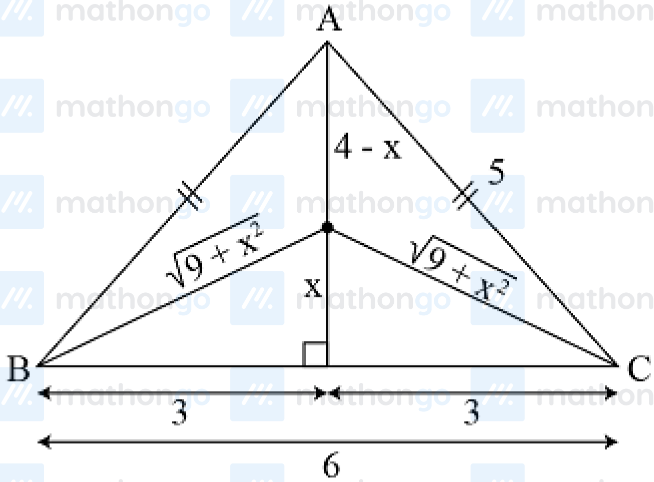
Q1 (1) - Single Correct

$L=4-x+2\sqrt{9+x^2}$, where $0\le x\le 4$

$\frac{dL}{dx}=\frac{2x}{\sqrt{9+x^2}}-1=0$

$\Rightarrow \quad 4x^2=9+x^2$

$\Rightarrow \quad x=\sqrt{3}$



Now, $L(0)=L(4)=10$

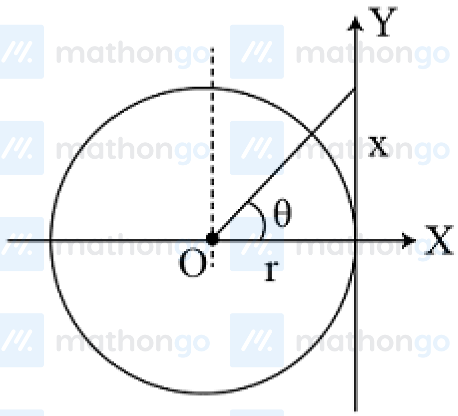
$L(\sqrt{3})=4-\sqrt{3}+2\cdot\sqrt{12}$

$\quad =4-\sqrt{3}+4\sqrt{3}=4+3\sqrt{3}$

$\therefore$ Minimum length of wire $=4+3\sqrt{3}$.

Q2 (2) - Single Correct

$\tan\theta=\frac{x}{r}$ or $x=r\tan\theta$



or $\dfrac{dx}{dt}=r\sec^2\theta\left(\dfrac{d\theta}{dr}\right)$

$\quad =r\omega\sec^2\theta=v\sec^2\theta$

[$\because v=r\omega$, where ω is angular velocity] where $\theta=\frac{2\pi}{8}$

$\therefore \quad \frac{dx}{dt}=v\sec^2\left(\frac{\pi}{4}\right)=2v=40$ km/h

Q3 (3) - Single Correct

Obviously, f is increasing and g is decreasing in R.

Hence, $\quad f\left[g\left(\alpha^2-2\alpha\right)\right]>f[g(3\alpha-4)]$

$\Rightarrow \quad g\left(\alpha^2-2\alpha\right)>g(3\alpha-4) \quad [\because f$ is increasing$]$

$\Rightarrow \quad \alpha^2-2\alpha<3\alpha-4 \quad [\because g$ is decreasing$]$

$\Rightarrow \quad \alpha^2-5\alpha+4<0$

$\Rightarrow \quad (\alpha-1)(\alpha-4)<0$

$\alpha\in(1,4)$

Q4 (3) - Single Correct

Since, $F(x)$ is the anti-derivative of $f(x)$

$\therefore \int f(x)dx=F(x)$

$\Rightarrow \quad F'(x)=f(x)$ and $F''(x)=f'(x)$

Now, $F'(x)=0\Rightarrow f(x)=0$

i.e. at $x=\frac{\pi}{2},\pi,\frac{3\pi}{2}$

At $x=\frac{\pi}{2}, F'(x)$ changes sign from positive to negative.

Hence, $y=F(x)$ has maxima at $x=\frac{\pi}{2}$ at $x=\pi$. $F'(x)$ does not change sign. $\Rightarrow$ Point of inflection at $x=\pi$.

At $x=\frac{3\pi}{2}$

$F(x)$ changes sign from negative to positive.

$\Rightarrow$ Minima at $\frac{3\pi}{2}$

Q5 (2) - Single Correct

We have,

$$f(x)=\frac{2}{3}\sqrt{x^3}-\frac{x}{2}+\int_1^x\left(\frac{1}{2}+\frac{1}{2}\cos 2t-\sqrt{t}\right)dt$$

$$f'(x)=\frac{2}{3}\times\frac{1}{2}\left(x^3\right)^{-1/2}3x^2-\frac{1}{2}+\left(\frac{1}{2}+\frac{1}{2}\cos 2x-\sqrt{x}\right)$$

$$=\left(x^3\right)^{-1/2}\cdot x^2-\frac{1}{2}+\left[\frac{1}{2}(1+\cos 2x)-\sqrt{x}\right]$$

$$=x^{1/2}-\frac{1}{2}+\frac{1}{2}(1+\cos 2x)-\sqrt{x}$$

$$=\frac{1}{2}\cos 2x$$

Put $f'(x)=0$, then

$\frac{1}{2}\cos 2x=0$

$\Rightarrow\cos 2x=0$

$\Rightarrow \quad 2x=(2n+1)\frac{\pi}{2}, n\in 1$

$\Rightarrow \quad x=(2n+1)\frac{\pi}{4}, n\in 1$

But $x>0$ $\quad$ [as $\sqrt{x}$ and $\sqrt{x^3}$ defined for $x>0$]

$\therefore x=\frac{\pi}{4},\frac{3\pi}{4},\frac{5\pi}{4},\frac{7\pi}{4}$

So, there are four critical points.

Q6 (1) - Single Correct

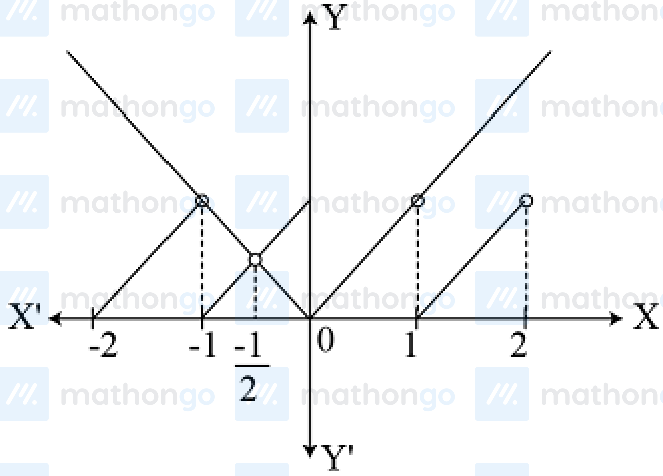
$\because f'(x)=|x|-\{x\}$

$\therefore \quad f(x)$ is decreasing.

$\therefore \quad f'(x)<0$

$|x|-\{x\}<0$

$|x|<\{x\}$



It is clear from the figure, $x \in (-1/2, 0)$

Q7 (2) - Single Correct

Slope of the given line $3x + 2y + 1 = 0$ is $-3/2$.

Let us locate the point on the curve at which the tangent is parallel to the given line.

On differentiating the curve both sides w.r.t. x, we get

$6x - 8y\frac{dy}{dx} = 0 \Rightarrow \left(\frac{dy}{dx}\right)_{(x_0, y_0)} = \frac{3x_0}{4y_0} = -\frac{3}{2}$

$[\because$ tangent is parallel to $3x + 2y = 1]$

Also, the point (x_0, y_0) lies on $3x^2 - 4y^2 = 72$

$3x_0^2 - 4y_0^2 = 72 \quad \Rightarrow \quad \frac{3x_0^2}{y_0^2} - 4 = \frac{72}{y_0^2}$

$\Rightarrow \quad 3(4) - 4 = \frac{72}{y_0^2} \quad \left[\text{as } \frac{x_0}{y_0} = -2\right]$

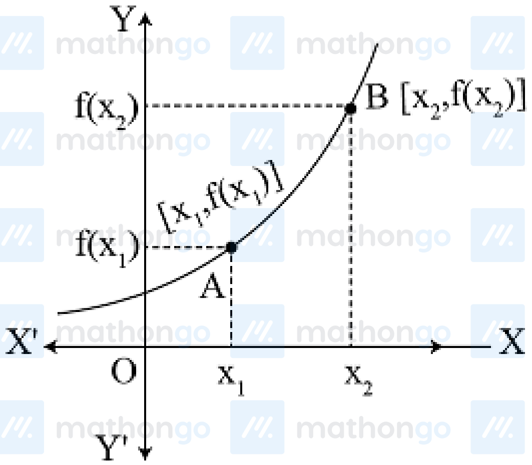
$\Rightarrow \quad y_0^2 = 9 \Rightarrow y_0 = \pm 3$

So, the required points are $(-6, 3)$ and $(6, -3)$. Distance of $(-6, 3)$ from the given line $= \frac{|-18+6+1|}{\sqrt{13}} = \frac{11}{\sqrt{13}}$ and distance of $(6, 3)$ from the given line. Thus, $(-6, 3)$ is the required point $= \frac{|18-6+1|}{\sqrt{13}} = \sqrt{13}$

$x_0 + y_0 = -6 + 3 = -3$

Q8 (3) - Single Correct

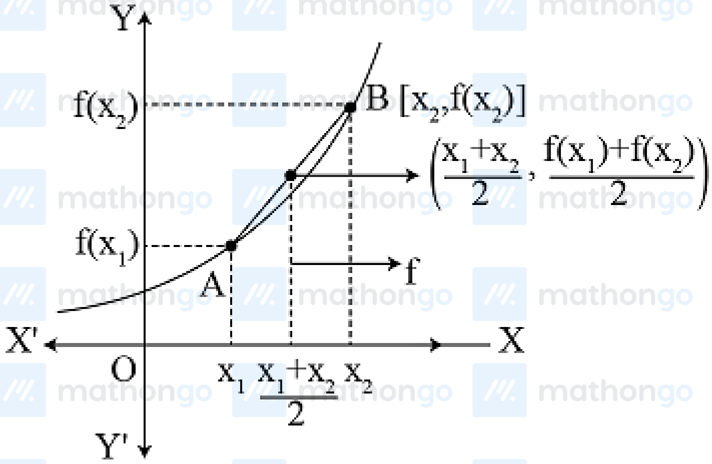
We know that. if $f'(x) > 0$ and $f^{\mu}(x) > 0$, then graphically it can be expressed as let $x_1 < x_2$



and $x_1 < \frac{x_1+x_2}{2} < x_2$

and $\frac{f(x_1)+f(x_2)}{2}$ is the mid-point of the chord joining A and B.

Thus, it can be expressed from the figure as



$\therefore \quad f\left(\frac{x_1+x_2}{2}\right) < \frac{f(x_1)+f(x_2)}{2}$

Q9 (3) - Single Correct

Given, $f'(\sin x) < 0, f''(\sin x) > 0, \forall x \in \left(0, \frac{\pi}{2}\right)$

$$g(x) = f(\sin x) + f(\cos x)$$

$$\Rightarrow g'(x) = f'(\sin x) \cdot \cos x + f'(\cos x)(-\sin x)$$

$$\Rightarrow g''(x) = -f'(\sin x)\sin x + \cos^2 x f''(\sin x)$$

$$+f''(\cos x)\sin^2 x - f'(\cos x)\cos x > 0, \forall x \in \left(0, \frac{\pi}{2}\right)$$

$$\left[\begin{array}{l}\text{as it is given, } f'(\sin x)=t'\left[\cos\left(\frac{\pi}{2}-x\right)\right]<0 \\ \text{and } f''(\sin x)=t''\left[\cos\left(\frac{\pi}{2}-x\right)\right]>0\end{array}\right]$$

$\Rightarrow \quad g'(x)$ is increasing in $\left(0, \frac{\pi}{2}\right)$ and $g \cdot\left(\frac{\pi}{4}\right)=0$

$\Rightarrow g'(x)>0, \forall x \in\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

and $g'(x)<0, \forall x \in\left(0, \frac{\pi}{4}\right)$

Q10 (1) - Single Correct

$$f'(x)=\begin{cases}1 & , x \geq 1 \\ -3 x^2, & x<1\end{cases}$$

$f(x)$ is increasing in $[1, \infty)$ and $f(x)$ is decreasing in $(-\infty, 1) \Rightarrow f(x)$ has minimum at $x=1$

$\therefore f\left(1^{-}\right) \geq f(1)$

$\Rightarrow [a+1]-1+1 \geq 0 \quad \Rightarrow \quad [a] \geq -1$

$\Rightarrow \quad a \geq -1$

Q11 (3) - Single Correct

$f(x)+f''(x)=-x|\sin x| \cdot f'(x)$

$f'(x) \cdot f(x)+f'(x) f''(x)=-x|\sin x| \cdot\left[f'(x)\right]^2$

$\Rightarrow \frac{1}{2} \cdot \frac{d}{d x}\left[\{f(x)\}^2+\left\{f'(x)\right\}^2\right]=-x|\sin x| \cdot\left[f'(x)\right]^2$

$\Rightarrow \frac{1}{2} \cdot \frac{d}{d x}[f(x)]=-x|\sin x| \cdot\left[f'(x)\right]^2$, as $x \geq 0$

$\Rightarrow F'(x)<0 \Rightarrow F(x)$ is decreasing function.

As, $x \geq 0 \Rightarrow F(x) \leq F(0)$

$\Rightarrow \quad[f(x)]^2+\left[f'(x)\right]^2 \leq[f(0)]^2+\left[f'(0)\right]^2$

$\Rightarrow \quad[f(x)]^2 \leq 25-\left[f'(x)\right]^2$

$\therefore$ Maximum value of $f(x)$ is 5.

Q12 (4) - Single Correct

Let $a<b$ and $f(x)=|x-a|+|x-b|, \forall x \in R$

So, $f(x)$ is decreasing in $(-\infty, a]$ constant is $[a, b]$ and increasing in $[b, \infty)$, we have

$f(0)=f(1)=f(-1) \Rightarrow\{-1,0,1\} \in[a, b]$

$\therefore|a-b|=2$

Q13 (3) - Single Correct

As, $\quad f(1)=0$

$f\left(1^{+}\right)>0$

$f\left(1^{-}\right)<0$

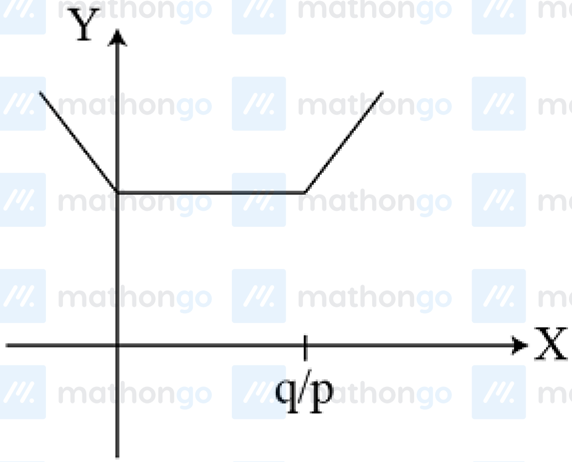
Hence, there are no local extremum at $x=1$.

Q14 (3) - Single Correct

$f(x)=|p x-q|+r|x|$

$=\left\{\begin{array}{ll}-p x+q-x, & x \leq 0 \\ -p x+q+x, & 0<x \leq q / p \\ p x-q+r x_1 & x>q / p\end{array}\right.$

For $r=p,\left\{\begin{array}{l}f^{\prime}(x)<0, \text { if } x \leq 0 \\ f^{\prime}(x)=0, \text { if } 0<x \leq q / p \\ f^{\prime}(x)>0, \text { if } x>q / p\end{array}\right.$



$\therefore$ Infinite points of minimum.

If $r>p$ or $r<p$, then only one point of minimum.

$\therefore$ When $r \neq p \Rightarrow$ one point of minimum.

Q15 (1) - Multiple Correct

$f(x)+f''(x)=-xg(x)f'(x)$

Let $h(x)=f^2(x)+[f'(x)]^2$

$\Rightarrow \quad h'(x)=2f(x)f'(x)+2f'(x)f''(x)$

$\quad\quad = 2f'(x)[-x]g(x)f'(x)=-2x\left[f'(x)\right]^2 g(x)$

$\therefore x=0$ is a point of maxima for $h(x)$.

Q16 (1, 2, 4) - Multiple Correct

$$f(x)=\begin{cases} x+2, & x<-1 \\ x^2, & -1\le x<1 \\ (x-2)^2, & x\ge 1 \end{cases}$$

At $x=-1, LHL=\lim_{h\to 0} f(-1-h)=\lim_{h\to 0}(-1-h+2)=1$

At $x=-1, RHL=\lim_{h\to 0} f(-1+h)=\lim_{h\to 0}(-1+h)^2=1$

Also, $f(-1)=(-1)^2=1$

$\therefore f(x)$ is continuous at $x=-1$

At $x=1, \mathrm{LHL}=\lim_{h\to 0} f(1-h)=(1-h)^2=1$

At $x=1, \mathrm{RHL}=\lim_{h\to 0} f(1+h)=\lim_{h\to 0}(1+h)^2=1$

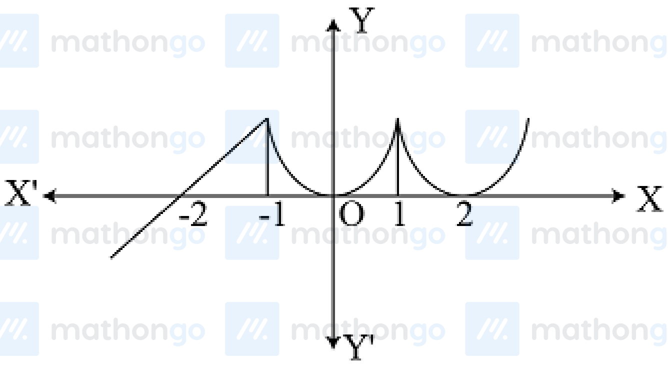
Also, $\quad f(1)=(1-2)^2=1$

$\therefore f(x)$ is continuous at $x=1$

$\therefore f(x)$ is continuous for all $x\in R$

$$f'(x)=\begin{cases} 1, & x<-1 \\ 2x, & -1\le x<1 \\ 2(x-2), & x\ge 1 \end{cases}$$

Clearly, $f'(x)$ changes sign at $x=-1,0,1,2$.



From the figure, it is clear that

$f(x)$ is not differentiable at $x=0,-1,1,2$

$\therefore f(x)$ has two local maxima and two local minima.

Q17 (1, 3, 4) - Multiple Correct

(1) Let $\quad f(x)=x-\tan^{-1} x$
$\Rightarrow \quad f'(x)=1-\frac{1}{1+x^2}$

$\Rightarrow \quad f'(x)=\frac{x^2}{1+x^2} \geq 0, \forall x \in(0,1)$

Then, $f(x)$ is increasing function. (as we know $x_1 \leq x_2 \Rightarrow f\left(x_1\right) \leq$ terms for increasing function)

$\therefore \quad x \geq 0, \forall x \in(0,1)$

$\Rightarrow \quad f(x) \geq f(0), \forall x \in(0,1)$
$\Rightarrow \quad x-\tan^{-1} x \geq 0-\tan^{-1}(0), \forall x \in(0,1)$

$\Rightarrow \quad x \geq \tan^{-1} x, \forall x \in(0,1)$
(2) Let $f(x)=\cos x-1+\frac{x^2}{2}$

$\Rightarrow f'(x)=-\sin x+x>0, \forall x \in(0,1)$

$f(x)$ is increasing in $(0,1)$
$\therefore f(x)>f(0)$

$\cos x-1+\frac{x^2}{2}>0$

$\Rightarrow \cos x>1-\frac{x^2}{2}$
(3) Let $f(x)=1+x \log \left(x+\sqrt{1+x^2}\right)-\sqrt{1+x^2}$

$\Rightarrow f'(x)=\log \left(x+\sqrt{x^2+1)}+\frac{x}{x+\sqrt{x^2+1}} \times x\left[1+\frac{x}{\sqrt{1+x^2}}\right]-\frac{x}{\sqrt{1+x^2}}\right.$

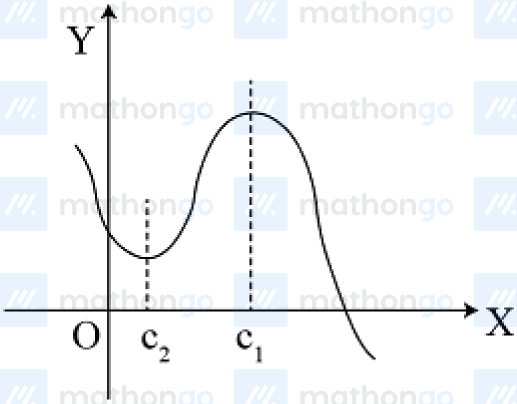
$=\log \left(x+\sqrt{x^2+1}\right)$

Q18 (1, 2, 3) - Multiple Correct

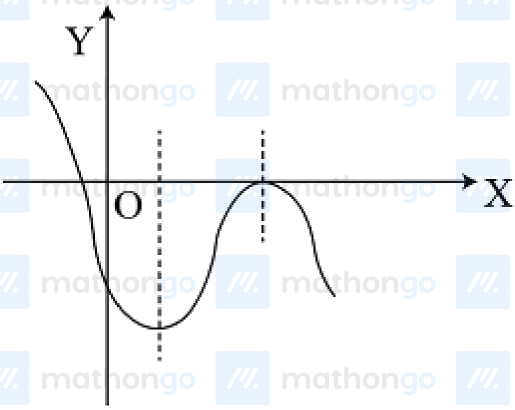
If $f'(x)=0$ has n real roots. $\Rightarrow (x)=0$ has atmost $(n+1)$ roots

If $(x)=0$ has 2 real roots, then.

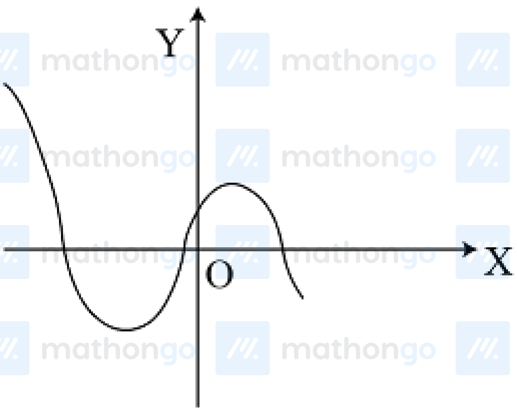
(1) $f(x)=0$ has 1 real root.



(2) $f(x) = o$ has 2 real roots.



(3) $f(x) = 0$ has 3 real roots.

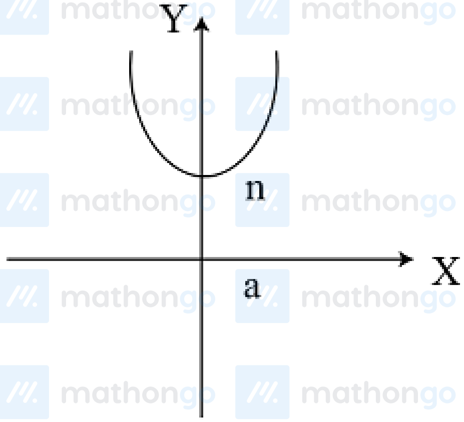


Q19 (1, 3) - Multiple Correct

Clearly, $\lim_{x \to a}[f(x)]$ is an integer and LHL and RHL should be same for existence of $\lim_{x \to a} f(x)$.

Let $\lim_{x \to a}[f(x)] = n(n \in l)$

Clearly, LHL and RHL both should be just greater than n as $f(x)$ is continuous at $x = a$.



$\therefore f(x)$ has local minimum at $x=a$.

Q20 (1, 2, 3) - Multiple Correct

(1) $f(x)=\begin{cases}1, & x=1\\ 0, & 0\le x<1\\ x, & -1\le x<0\end{cases}\Rightarrow$ not differentiable at $x=0$ in $(-1,1)$

(2) $f(0)=0$ and $\lim_{x\to 0}\frac{\tan x}{x}=1\Rightarrow$ not continuous at $x=0$

(3) $f(x)=\begin{cases}1, & x\notin \text{ integer}\\ 0, & x\in \text{ integer}\end{cases}\Rightarrow$ not continuous at $x=0$

(4) $f(x)=\begin{cases}x-\sin x, & 0\le x\le 1\\ -x+\sin x, & -1\le x<0\end{cases}\Rightarrow$ continuous and differentiable at $x=0$

Q21 (1, 2) - Multiple Correct

Given, $f(x)=1+x\log\left(x+\sqrt{x^2+1}\right)-\sqrt{1+x^2}$

$h(x)=f(x)-f^2(x)+f^3(x)$

$$\begin{aligned}\Rightarrow\quad h'(x)&=f'(x)-2f(x)f'(x)+3f^2(x)f'(x)\\&=f'(x)\left[1-2f(x)+3f^2(x)\right]\end{aligned}$$

We know that, $1+x\log\left(x+\sqrt{x^2+1}\right)-\sqrt{1+x^2}>0$

$\Rightarrow\quad f(x)>0$

Now, $1-2f(x)+3f^2(x)>0,\forall x\in(0,\infty)$

and $1-2f(x)+3f^2(x)<0,\forall x\in(-\infty,0)$

Hence, $h(x)$ is increasing in $(0,\infty)$ and decreasing in $(-\infty,0)$.

Q22 (3, 4) - Multiple Correct

Given, $f(x) = a(x - x_1)(x - x_2)$

$f(x) = a(x + x_2)(x - x_2)$

$f(x) = a(x^2 - x_2^2) \quad [\because x_1 + x_2 = 0]$

$\Rightarrow \quad f'(x) = a(2x - 2x_2)$

$f'(x) = 2a(x - x_2)$

$f''(x) = 2a > 0$

$\Rightarrow f(x)$ has an extremum.

Now, $f'(x) = 2a(x + x_1) \quad [\because x_1 + x_2 = 0]$

From this expression, we cannot say that

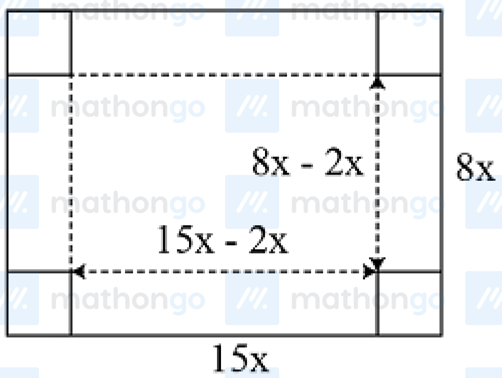
$f'(x) > 0$ or $f'(x) < 0, \forall x \in R$

So, $f(x)$ is non-monotonic on R.

Q23 (1, 3) - Multiple Correct

Here $I = 15x - 2a, b = 8x - 2a$ and $h = a$

$\therefore \quad$ Volume $= (8x - 2a)(15x - 2a)$



$V = 2a \cdot (4x - a)(15x - 2a) \ldots (i)$

$\frac{dV}{du} = 6a^2 - 46ax + 60x^2$

$\frac{d^2V}{dx^2} = 12a - 46x$

Here, $\left(\frac{\partial V}{da}\right) = 0 \Rightarrow 6x^2 - 23x + 15 = 0$

At $a = 5, \quad x = 3, 5/6$

$\frac{d^2V}{da^2} = 2(30 - 23)x$

At $x = 3, \quad \frac{d^2V}{da^2} = 2(30 - 69) < 0$

$\therefore$ Maximum at $x=3$, also at $x=5/6 \Rightarrow \frac{d^2V}{da^2}=0$

$\therefore$ At $x=5/6, V$ is minimum. Thus. sides are $8x=24$ and $15x=45$

Q24 (2, 3) - Multiple Correct

Since, $f(x)$ has local maxima at $x=-1$ and $f'(x)$ has local minima at $x=0$.

$\therefore f''(x)=\lambda x$

On integrating, we get

$$f'(x)=\frac{\lambda x^2}{2}+c \qquad [\because f'(-1)=0]$$

$$\Rightarrow \frac{\lambda}{2}+c \qquad =0 \Rightarrow \lambda=-2c \qquad \ldots \text{(i)}$$

Again, on integrating both sides, we get

$f(x)=\frac{\lambda x^3}{6}+cx+d$

$f(2)=\lambda(8/6)+2c+d=18$

and $f(1)=\frac{\lambda}{6}+c+d=-1$

From Eqs. (i), (ii) and (iii), we get

$f(x)=\frac{1}{4}\left(19x^3-57x+34\right)$

$\Rightarrow \quad f'(x)=\frac{1}{4}\left(57x^2-57\right)=\frac{57}{4}(x-1)(x+1)$

For maxima or minima, put $f'(x)=0 \Rightarrow x=1,-1$

Now, $\quad f''(x)=\frac{1}{4}(114x)$

At $x=1, \quad f''(x)>0$, minima

At $x=-1, \quad f''(x)<0$, maxima

$\therefore f(x)$ is increasing for $[1,2\sqrt{5}]. \Rightarrow \quad f(x)$ has local maxima at $x=-1$ and $\quad f(x)$ has local minima at $x=1$. Also,

$f(0)=34/4$

Q25 (2, 3, 4) - Multiple Correct

We have, $\quad f(x)=x\cos\frac{1}{x}, x\geq 1$

$f'(x)=\cos\frac{1}{x}+\frac{1}{x}\sin\frac{1}{x}$

$f''(x)=-\frac{1}{x^3}\cos\frac{1}{x}$

$\lim_{x\to\infty} f'(x)=1 \quad$ and $\quad f''(x)<0, \forall x\geq 1$

$\Rightarrow f'(x)$ is strictly decreasing in the interval $[1, \infty)$ Let $g(x) = f(x+2) - f(x)$

then $g'(x) = f'(x+2) - f'(x) < 0$ [$\because f'(x)$ is strictly decreasing]

$\Rightarrow g(x)$ is strictly decreasing on $(1, \infty)$

Now, $\lim_{x\to\infty} g(x) = \lim_{x\to\infty} f(x+2) - f(x)$

$= \lim_{x\to\infty}(x+2)\cos\frac{1}{x+2} - x\cos\frac{1}{x}$

$= \lim_{x\to\infty} x\left(\cos\frac{1}{x+2} - \cos\frac{1}{x}\right) + 2\cos\frac{1}{x+2}$

$= \lim_{x\to\infty} 2x\sin\frac{x+1}{x^2+2x}\sin\frac{1}{x^2+2x} + 2\cos\frac{1}{x+2}$

$= 2\lim_{x\to\infty}\frac{\sin\left(\frac{x+1}{x^2+2x}\right)}{\frac{x+1}{x^2+2x}} \times \frac{\sin\left(\frac{1}{x^2+2x}\right)}{\frac{1}{x^2+2x}} \times \frac{x(x+1)}{(x^2+2x)^2} + 2\cos\frac{1}{x+2}$

$= 2\times1\times1\times0+2=2$

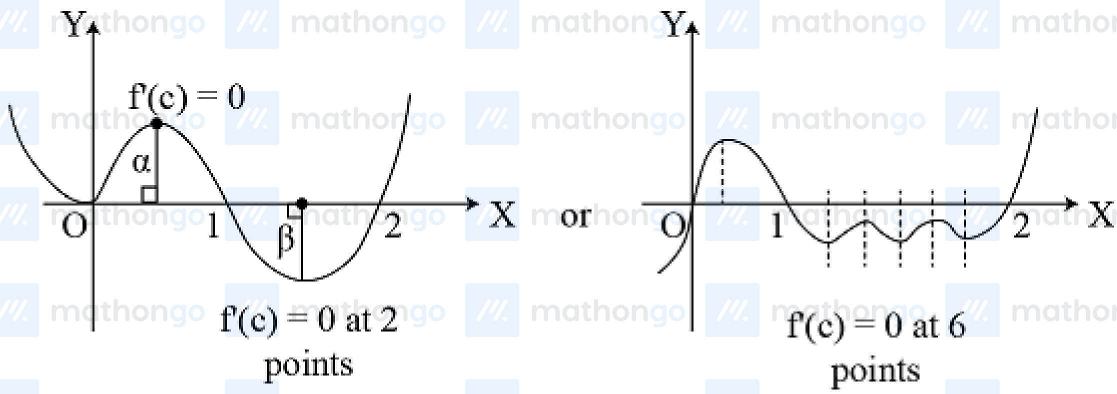
As $g(x)$ is decreasing on $[1, \infty)$ and $\lim_{x\to\infty} g(x) = 2$

$\therefore \quad g(x) > 2, \forall x \in [1, \infty)$

$\Rightarrow \quad f(x+2) - f(x) > 2, \forall x \in [1, 0)$

Q26 (2, 4) - Multiple Correct

$f(0) = f(1) = f(2) = 0$



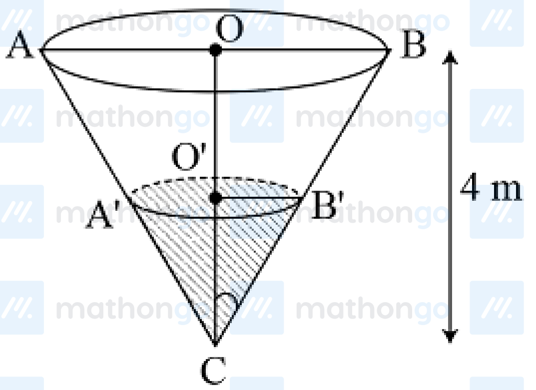
As, $f(x)$ is continuous and differentiable between $x=0$ and $x=2$.

$\therefore f'(c) = 0$ for two points $x = \alpha, \beta$. $\Rightarrow f'(c) = 0$ for atleast two $c \in (0, 2)$

Q27 (2) - Multiple Correct

Let α be the semi-vertical angle of the cone $C'AB$ whose height CO is 4 m and radius OB $= 2$ m, then

$\tan\alpha = \frac{1}{2}$



Let V be the volume of water in the conei.e. the volume of the cone

$CA'B'$

After timet min, h be the height of the water, then

$V=\frac{1}{3}\pi(O'B')^2\,(CO')$

$\Rightarrow V=\frac{1}{3}\pi h^3\tan^2\alpha \quad \left[\begin{array}{l}\because \tan\alpha=\frac{O'B'}{CO'}=\frac{O'B'}{h}\\ \Rightarrow O'B'=h\tan\alpha\end{array}\right]$

$\Rightarrow V=\frac{\pi}{12}h^3$

Now, $\frac{dV}{dt}=\frac{\pi}{12}3h^2\frac{dh}{dt}=\frac{\pi}{4}h^2\frac{dh}{dt}$

$\Rightarrow \quad 77\times10^3=\frac{\pi h^2}{4}\cdot\frac{dh}{dt} \quad \left[\begin{array}{l}\because 1\text{ L}=10^3\text{ cm}^3\\ \therefore 77\text{ L}=77\times10^3\text{ cm}^3\\ \text{i.e. }\frac{\text{dV}}{\text{dt}}=77\text{ L}=77\times10^3\text{ cm}^3\end{array}\right]$

$\Rightarrow \frac{\text{dh}}{dt}=\frac{77\times4\times10^3}{\pi h^2}$

$\therefore \quad \left(\frac{\text{dh}}{dt}\right)_h=\frac{77\times4\times10^3}{\pi\times(70)^2}=\frac{77\times4\times7\times10^3}{22\times70\times70}=20\text{ cm/min}$

Q28 (1, 2) - Multiple Correct

Given, $\sqrt{xy}=a+x$

$\Rightarrow \frac{1}{2\sqrt{xy}}(y+xy')=1$

Since, tangent cuts-off equal intercept from the axes.

$y'=-1$

$\Rightarrow y-x=2\sqrt{xy}=2(a+x)$

$\Rightarrow y=2a+3x\Rightarrow x(2a+3x)=(a+x)^2$

$\Rightarrow 2ax+3x^2=a^2+x^2+2ax$

$\Rightarrow 2x^2=a^2$

$\therefore x=\pm a/\sqrt{2}$

Q29 (3) - Paragraph 1

Explanation (For Question 29, 30)

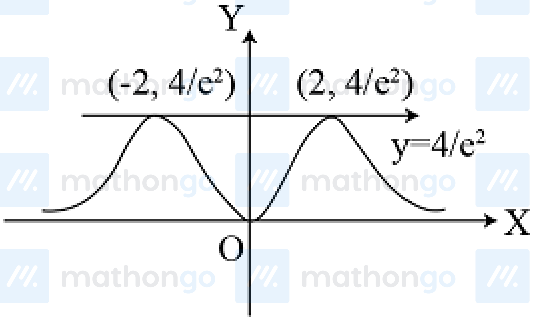
We have, $f(x)=x^2e^{-|x|}=\begin{cases}x^2e^{-x}, & x\geq 0\\ x^2e^{x}, & x<0\end{cases}$

$\Rightarrow \quad f'(x)=\begin{cases}x\cdot e^{-x}\cdot(2-x), & x\geq 0\\ x\cdot e^{x}\cdot(x+2), & x<0\end{cases}$

$f(x)$ is increasing in $(-\infty,-2)\cup(0,2)$ and $f(x)$ is decreasing in $(-2,0)\cup(2,\infty)$.

Now, $\lim_{x\to+\infty}f(x)=0$ and $\lim_{x\to\pm 2}f(x)=\frac{4}{e^2}f(0)=0$

$\therefore$ Graph of $f(x)$ will be as shown in the figure given below



$y=kx^2$ will intersect $y=e^{|x|}$,

then, $kx^2=e^{|x|}\Rightarrow x^2e^{-|x|}=\frac{1}{k}$

So, $y=x^2e^{-|x|}$ will be cut by line $y=\frac{1}{k}$ At exactly two points, if $\frac{1}{k}=\frac{4}{e^2}$

$\Rightarrow \quad k=\frac{e^2}{4}$

$\therefore$ Area $=2\int_0^2 x^2e^{-x}dx$

$=2\left\{\left[-x^2e^{-x}\right]_0^2+\int_0^2 2xe^{-x}dx\right\}$

$=2\left\{-\frac{4}{e^2}+2[-xe^{-x}]_0^2-2[e^{-x}]_0^2\right\}=2\left(2-\frac{10}{e^2}\right)$

From the above figure

Range $=\left[0,\frac{4}{e^2}\right]$

Q30 (1) - Paragraph 1

Q31 (4) - Paragraph 2

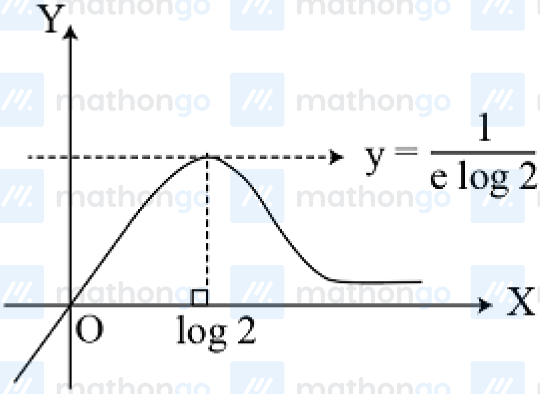
Explanation (For Question 31, 32, 33)

We have, $f(x)=x\cdot 2^{-x}$

$\therefore f'(x) = 2^{-x}(1 - x\log 2)$

and $f'(x) = 2^{-x}\log 2(x\log 2 - 2)$

Clearly, $f(x)$ is increasing in $\left(-\infty, \frac{1}{\log 2}\right)$ and decreasing in $\left(\frac{1}{\log 2}, \infty\right)$



Clearly, $f(x) = k$ has two distinct roots for $k \in \left(0, \frac{1}{e\log 2}\right)$.

Given, $f(x) = x \cdot 2^{-x}$ and $g(x) = \max\{f(t) : x \le t \le x+1\}$

As, $f(x)$ is increasing in $\left(-\infty, \frac{1}{\log 2}\right)$, hence the maximum value of $g(x)$ occurs at $t = x+1 \therefore \quad g(x) = f(x+1) = (x+1)\cdot 2^{-(x+1)}$

Q32 (2) - Paragraph 2

Q33 (4) - Paragraph 2

Q34 (4) - Paragraph 3

Explanation (For Question 34, 35)

$$\lim_{x\to 0}\frac{1}{x}\log\begin{vmatrix}\frac{f(x)}{x} & 1 & 0\\ 0 & \frac{1}{x} & 1\\ 1 & 0 & \frac{1}{x}\end{vmatrix} = 2$$

$$\Rightarrow \quad \lim_{x\to 0}\frac{\log\left[\frac{f(x)}{x^3}+1\right]}{x} = 2 \text{ ...(i)}$$

For limit to exist, $\lim_{x\to 0}\frac{f(x)}{x^3} = 0$

$f(x) = a_0x^6 + a_1x^5 + a_2x^4$

Also, $\quad f'(0) = f'(2) = f'(1) = 0$

$$\Rightarrow \quad f'(x) = 6a_0x^5 + 5a_1x^4 + 4a_2x^3$$
$$= x^3\left(4a_2 + 5a_1x + 6a_0x^2\right)$$
$$f'(2) = 0 \Rightarrow 24a_0 + 10a_1 + 4a_2 = 0 \quad \ldots(\text{ii})$$
$$f'(1) = 0 \Rightarrow 6a_0 + 5a_1 + 4a_2 = 0 \quad \ldots(\text{ii})$$

From Eq. (i), we get

$$\log\left\{\lim_{x\to 0}\left(\frac{f(x)}{x^3} + 1\right)^{1/x}\right\} = 2 \quad \Rightarrow \log\left\{e^{\lim_{x\to 0}\frac{f(x)}{x^4}}\right\} = 2$$

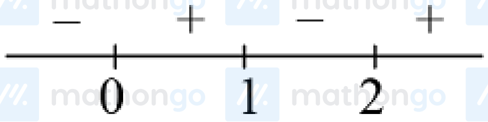
$$\Rightarrow \quad \lim_{x\to 0}\frac{a_0x^6 + a_1x^5 + a_2x^4}{x^4} = 2$$

$\Rightarrow \quad a_2 = 2$ ...(iii)

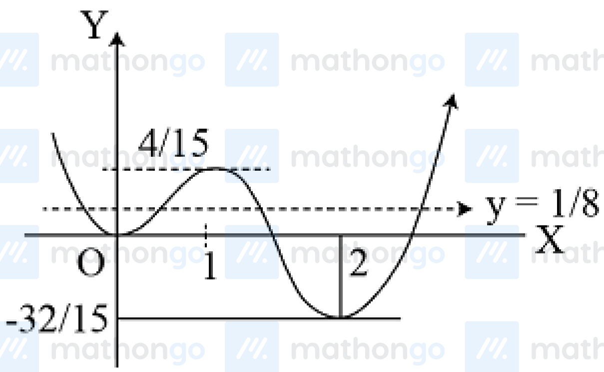
From Eqs. (ii), (iii) and (iv), we get $\Rightarrow \quad a_1 = -\frac{12}{5}, a_0 = \frac{2}{3}$

$\therefore \quad f(x) = \frac{2}{3}x^6 - \frac{12}{5}x^5 + 2x^4$ ...(iv)

$$\therefore \quad f(x) = \frac{2}{3}x^6 - \frac{12}{5}x^5 + 2x^4$$
$$f'(x) = 4x^5 - 12x^4 + 8x^3$$
$$= 4x^3 \cdot (x-1)(x-2)$$



Graph for $f(x)$ is



Q35 (1) - Paragraph 3

Q36 (4) - Integer Type

Using LMVT for f in $[1, 2]$,

$\forall c \in (1, 2), \frac{f(2) - f(1)}{2 - 1} = f'(c) \leq 2$

$f(2) - f(1) \leq 2$

$\Rightarrow \quad f(2) \leq 4$ ...(i)

Again, using LMVT in $[2,4]$,

$\forall d \in (2,4), \frac{f(4)-f(2)}{4-2} = f'(d) \leq 2$

$\therefore f(4) - f(2) \leq 4$

$8 - f(2) \leq 4$

$f(2) \geq 4 \quad \ldots$(ii)

From Eqs. (i) and (ii), we get

$f(2) = 4$

Q37 (5) - Integer Type

Given, $f''(x) - g''(x) = 0$

On integrating both sides, we get

$\Rightarrow \quad f'(x) - g'(c) = c,$

At $x = 1, \quad f'(1) - g'(1) = c$

$\Rightarrow \quad c = -2$

$\therefore f'(x) - g'(x) = -2$

Again integrating both sides, we get

$f(x) - g(x) = -2x + c_1$

At $x = 2, \quad f(2) - g(2) = -4 + c$

$\therefore c_1 = -2$

or $f(x) - g(x) = -2x - 2$

$\therefore \left|f\left(\frac{3}{2}\right) - g\left(\frac{3}{2}\right)\right| = |-3-2| = 5$

Q38 (9) - Integer Type

Since, $x + 4y = 14$ is normal to the curve $y^2 = \alpha x^3 - \beta$ at $(2,3)$.

$\therefore$ Slope of $(x + 4y = 14) = \left(\frac{dy}{dx}\right)_{(2,3)}$

$\Rightarrow \quad -\frac{1}{4} = \left[-\frac{1}{\left(3\alpha\frac{x^2}{2y}\right)}\right]_{(2,3)}$

$\Rightarrow \quad -\frac{1}{4} = -\frac{1}{2\alpha} \Rightarrow \alpha = 2$

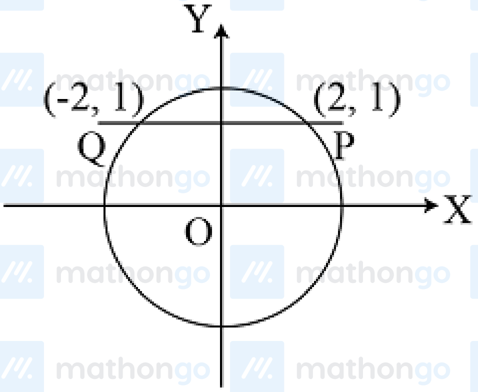
From Eq. (i), we have

$$(3)^2 = 2(2)^3 - \beta$$
$$\Rightarrow 9 = 16 - \beta$$
$$\Rightarrow \beta = 7$$
$$\therefore \alpha + \beta = 2 + 7 = 9$$

Q39 (2) - Integer Type

Here, $1 \leq |\sin x| + |\cos x| \leq \sqrt{2}$

$$\therefore \quad \left[\begin{array}{l} |\sin x| + |\cos x|] = 1 \\ x^2 + 1 = 5 \Rightarrow x = \pm 2 \end{array} \right.$$



$\therefore P(2, 1)$ and $Q(-2, 1)$

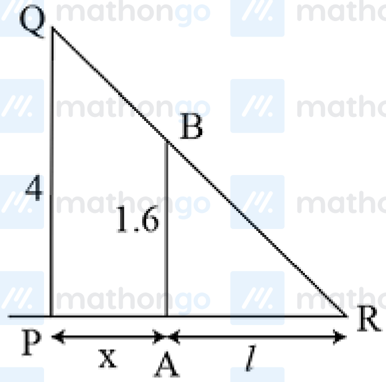
Slope of tangent of the curve at P is 2 and slope of tangent of the curve at Q is -2.

$\therefore$ Angle of intersection is

$$\tan^{-1}(2) = \tan^{-1}(k)$$
$$\therefore k = 2$$

Q40 (5) - Integer Type

Let $PQ = 4$ m be the height of the pole and $AB = 1.6$ m be the height of the man. Let the end of shadow be R and it is at a distance of l from A when the man is at distance x from PQ at some instant.



We have, $\frac{PQ}{AB} = \frac{PR}{AR}$ [$\because \Delta PQR$ and $\triangle ABR$ are similar]

$\Rightarrow \frac{4}{1.6} = \frac{x+l}{l} \Rightarrow 2x = 3l$

$\Rightarrow \quad 2\frac{dx}{dt} = 3\frac{dl}{dt}$

$\Rightarrow \quad \frac{dl}{dt} = \frac{2}{3} \times 30 \text{ m/min} = 20 \text{ m/min}$

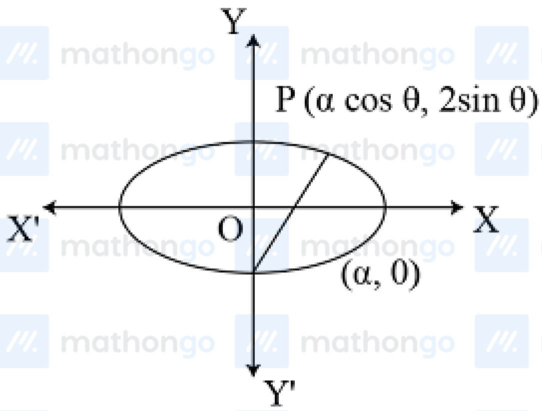
But $\frac{dl}{dt} = k \Rightarrow \frac{k}{4} = \frac{20}{4} = 5$

Q41 (3) - Integer Type

The equation of the given curve can be expressed as shown in the figure given below

$\frac{x^2}{\alpha^2} + \frac{y^2}{4} = 1, \text{ where } 4 < \alpha^2 < 8$

which is equation of the ellipse.



Let us consider a point $P(\alpha \cos\theta, 2\sin\theta)$ on the ellipse.

Let the distance of $P(\alpha\cos\theta, 2\sin\theta)$ from $(0, -2)$ is L, then

$$L^2 = (\alpha\cos\theta - 0)^2 + (2\sin\theta + 2)^2$$

$$\frac{dL^2}{d\theta} = \cos\theta\left[-2\alpha^2\sin\theta + 8\sin\theta + 8\right]$$

Now, $\frac{dL^2}{d\theta} = 0$

$\Rightarrow \quad \cos\theta\left[-2\alpha^2\sin\theta+8\sin\theta+8\right]=0$

Either $\cos\theta=0$ or $\left(8-2\alpha^2\right)\sin\theta+8=0$

$\Rightarrow\theta=\frac{\pi}{2}$ or $\sin\theta=\frac{4}{\alpha^2-4}$

Since, $\alpha^2<8\Rightarrow\alpha^2-4<4$

$\Rightarrow\frac{4}{\alpha^2-4}>1$

$\therefore \quad \sin\theta>1$, which is not possible.

$$\frac{d^2L^2}{d\theta^2}=\cos\theta\left[-2\alpha^2\cos\theta+8\cos\theta\right]$$

Further, $\quad +(-\sin\theta)\left[-2\alpha^2\sin\theta+8\sin\theta+8\right]$

$$\text{At } \theta=\frac{\pi}{2},\frac{d^2L^2}{d\theta^2}=0-\left(16-2\alpha^2\right)=2\left(\alpha^2-8\right)<0 \quad \left[\text{as } \alpha^2<8\right]$$

So, L is maximum at $\theta=\frac{\pi}{2}$ and the farthest point is $(0,2)$.

The reflection of point $P(0,2)$ in X -axis is $R(0,-2)$. Now, distance of $R(0,-2)$ from the line

$3x-4y+7=0$,

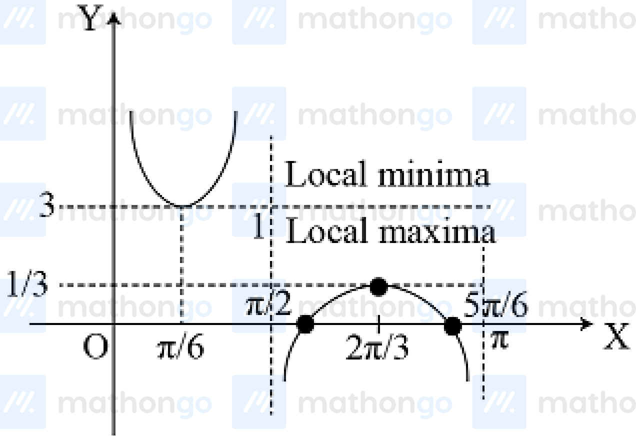
$D=\left|\frac{3\times0-4(-2)+7}{\sqrt{(3)^2+(-4)^2}}\right|=\left|\frac{15}{5}\right|=3$

Q42 (3) - Integer Type

Here, $y=\frac{2\sin\left(x+\frac{\pi}{6}\right)\cos x}{2\sin x\cdot\cos\left(x+\frac{\pi}{6}\right)}=\frac{\sin\left(2x+\frac{\pi}{6}\right)+\sin\frac{\pi}{6}}{\sin\left(2x+\frac{\pi}{6}\right)-\sin\frac{\pi}{6}}$

$y=1+\frac{1}{\sin\left(2x+\frac{\pi}{6}\right)-\sin\frac{\pi}{6}}$

y is minimum, if $2x+\frac{\pi}{6}=\frac{\pi}{2}$

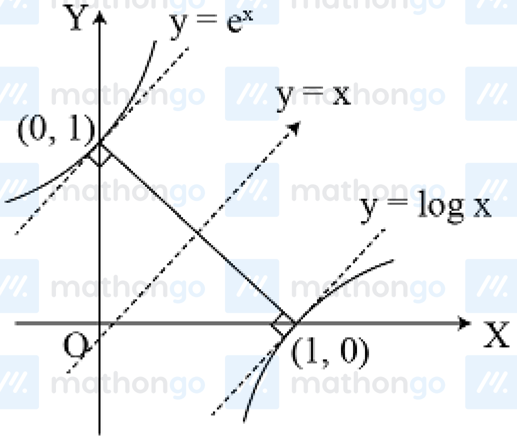


$\therefore \quad x=\frac{\pi}{6}$

$\Rightarrow \quad y_{min}=1+2=3$

Q43 (2) - Integer Type

Minimum distance is the distance along the common normal to both the curves i.e. $y = x$ must be parallel to the tangent as both the curves are inverse of each other.



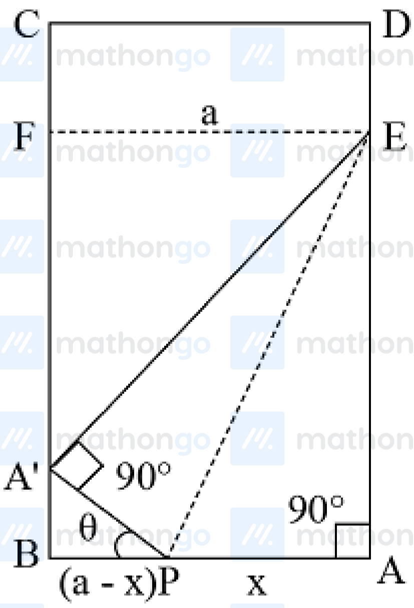
$\left(\frac{dy}{dx}\right)_{x_1} = e^{x_1} = 1 \Rightarrow x_1 = 0$ and $y_1 = 1$

$\therefore \quad AB = \sqrt{2} = \sqrt{k}$

$\Rightarrow \quad k = 2$

Q44 (5) - Integer Type

Let the corner A of leaf $ABCD$ be folded over to A which is the inner edge BC of the page.



Let $AP = x$, then $\quad AB = a$

$\therefore \quad BP = a - x$

If $\angle A'PB = \theta B_{a-x}$, then

$\angle EA'F = \theta$

Since, EF is parallel to AB, then $EF = a$

in $\Delta A'BP, \cos\theta = \frac{BP}{PA'} \Rightarrow \frac{a-x}{x}$

In $\Delta A'FE, A'E = EF$

$\theta = \frac{a}{\sqrt{1-\cos^2\theta}}$

$= \frac{a}{\sqrt{1-\left(\frac{a-x}{x}\right)^2}} = \frac{ax}{\sqrt{x^2-(a-x)^2}} = \frac{ax}{\sqrt{2ax-a^2}} = AE$

Now, ΔAPE is folded to $A'PE =$ Area of folded part

$=$ Area of $\Delta PAE = \frac{1}{2}AP \cdot AE$

$= \frac{1}{2} \times \frac{ax}{\sqrt{2ax-a^2}} = A[\text{ say }]$

$A^2 = \frac{a^2x^4}{4(2ax-a^2)} = \frac{a^2/4}{\left(\frac{2a}{x^3}-\frac{a^2}{x^4}\right)} = \frac{a^2/4}{y}$

where, $y = \frac{2a}{x^3} - \frac{a^2}{x^4}; 0 < x < a$

Obviously, $A^2(1 \in A)$ is minimum, when y is maximum.

There, $y = \frac{2a}{x^3} - \frac{a^2}{x^4} \Rightarrow \frac{dy}{dx} = -\frac{6a}{x^4} + \frac{4a^2}{x^5}$

and $\frac{d^2y}{dx^2} = \frac{24a}{x^5} - \frac{20a^2}{x^6}$

For maxima or minima of y_1

$$\frac{dy}{dx} = -\frac{6a}{x^4} + \frac{4a^2}{x^5} = 0$$
$$x = 2a/3$$

$$\text{When } x = \frac{2a}{3}, \text{ then } \frac{d^2y}{dx^2} = \frac{4a}{x^5}\left(6 - \frac{5a}{x}\right)$$
$$= 4a(3/2a)^5\left(6 - 5a\frac{3}{2a}\right)$$
$$= -4a(3/2a)^5 \cdot \frac{3}{2} < 0$$

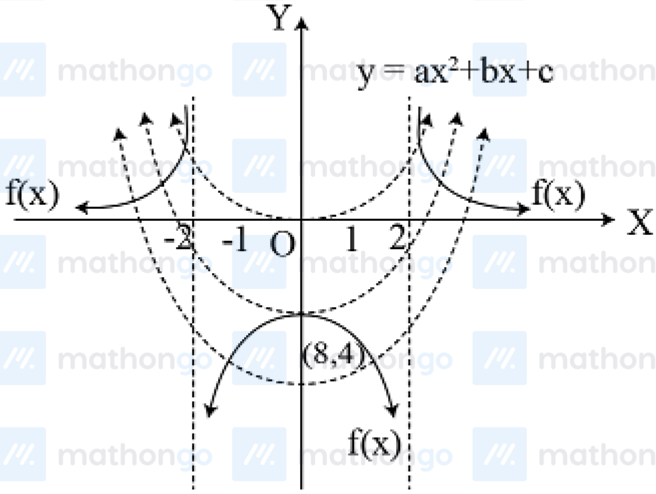
$\Rightarrow y$ is maximum i.e. A is minimum when $x = \frac{2a}{3}$, which is the only critical point (least), So, the folded area is least, when $\frac{2}{3}$ of the width of the page in folded over.

$\Rightarrow \frac{p}{q} = \frac{2}{3}$

$\therefore p + q = 5$

Q45 (2) - Integer Type

$f(x)=\frac{1}{x^2-4}$ is shown as below



From the above figure, the minimum number of point of intersection is 2

Q46 (6) - Integer Type

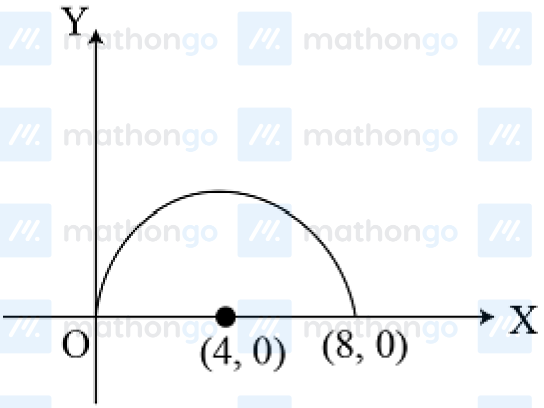
We have,

$f(x)=\sqrt{16-(x-4)^2}-\sqrt{1-(x-7)^2}$

Now, consider $\quad y=\sqrt{16-(x-4)^2}$

$\Rightarrow \quad (x-4)^2+y^2=16, y>0$

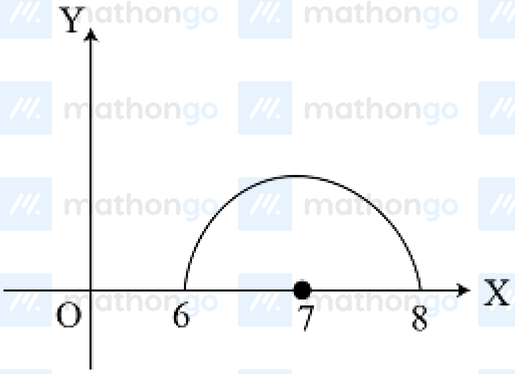
is a semi-circle with centre $(4,0)$ and radius 4



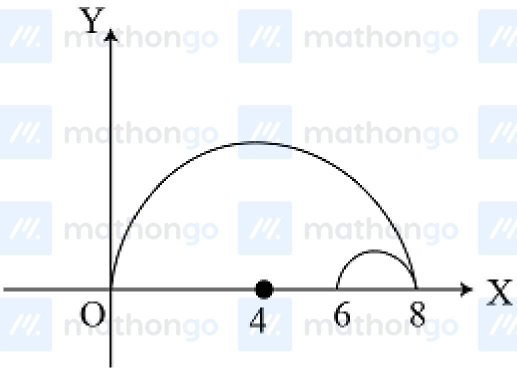
Similarly, $y=\sqrt{1-(x-7)^2}$

$\Rightarrow \quad (x-7)^2+y^2=1, y>0$

is a semi-circle with centre $(7,0)$ and radius 1.



On combining the above two figures, we have



$[f(x)]_{\max}$ = Maximum vertical distance between the two curves which occurs when $x = 6$.

$\therefore [f(x)]_{\ln ax} = \sqrt{16-(6-4)^2} - 0$

$= \sqrt{12} = \sqrt{k}$

$\therefore k = 12 = 2^2 \times 3^1$

So, the number of divisors is 6.